

Octahedral connectivity reconfigures interfacial carrier-selective properties for efficient perovskite solar cells

Received: 3 January 2026

Accepted: 28 July 2026

Published online: 03 September 2026

 Check for updates

Yalan Zhang^{1,7}, Zheng Liang^{1,7}, Seong Chan Cho^{1,2}, Seong-Ho Cho¹, Guiming Fu¹, Xin Chen³, Boyuan Liu⁴, Sang-Uk Lee¹, Jae Hun Seol², Sanwan Liu¹, Hui Zhang⁴, Kui Zhao³, Sang Uck Lee²✉, Xu Pan^{4,5}✉ & Nam-Gyu Park^{1,6}✉

Regulation of electron- or hole-selective interfacial properties has traditionally relied on extrinsic chemical doping. Here we report a structurally driven strategy for modulating carrier-selective electronic characteristics through the organic-cation-induced reconfiguration of lead iodide octahedral connectivity. Two closely related imidazoline-based cations, which differ by only a single heteroatom, drive the formation of one-dimensional organic lead triiodide phases with distinct octahedral-sharing patterns. Single-crystal analysis combined with theoretical calculations reveals that the variation in connectivity reshapes orbital coupling, leading to pronounced changes in the work function and absolute band-edge positions that give rise to distinct carrier-selective interfacial behaviour without the introduction of extrinsic dopants. Leveraging this structural reconfiguration-induced electronic modulation, we realize complementary charge-selective contacts in perovskite solar cells, achieving a power conversion efficiency of 27.61% (certified steady-state 27.19%), together with excellent operational stability and scalability, including 22.26% efficiency in 655-cm² modules.

Interfaces in perovskite solar cells (PSCs) play a crucial role in governing charge extraction and ensuring operational stability. Within the layered device architecture, the perovskite absorber is confined by two semiconducting interfaces that dictate carrier selectivity and overall band alignment^{1–3}. Extensive interface engineering strategies, including chemical passivation^{4,5}, dipole alignment^{6,7} and work function modulation^{8,9}, have substantially enhanced device performance by mitigating interfacial defects and improving band alignment. Never-

theless, most of these approaches have focused on exploring modifier molecules that act at either a p-type or n-type single-layer interface^{10–14}, leaving the opposite contact electronically unbalanced. Among them, one particularly effective approach has been to treat the perovskite absorber with bulky organic salts, where these species often crystallized into secondary phases, low-dimensional perovskites or related perovskitoids that compensated band-edge offsets and reinforced built-in electric fields^{15–19}.

¹School of Chemical Engineering and Center for Antibonding Regulated Crystals, Sungkyunkwan University, Suwon, Republic of Korea. ²School of Chemical Engineering, Sungkyunkwan University, Suwon, Republic of Korea. ³Key Laboratory of Applied Surface and Colloid Chemistry, National Ministry of Education, Shaanxi Key Laboratory for Advanced Energy Devices, Shaanxi Engineering Lab for Advanced Energy Technology and School of Materials Science and Engineering, Shaanxi Normal University, Xi'an, People's Republic of China. ⁴University of Science and Technology of China, Hefei, People's Republic of China. ⁵School of Microelectronics, Hefei University of Technology, Hefei, People's Republic of China. ⁶SKKU National Lab for Intelligent Energy Solution Technology (SIEST), Sungkyunkwan University, Suwon, Republic of Korea. ⁷These authors contributed equally: Yalan Zhang, Zheng Liang. ✉e-mail: suleechem@skku.edu; xpan@rntek.cas.cn; npark@skku.edu

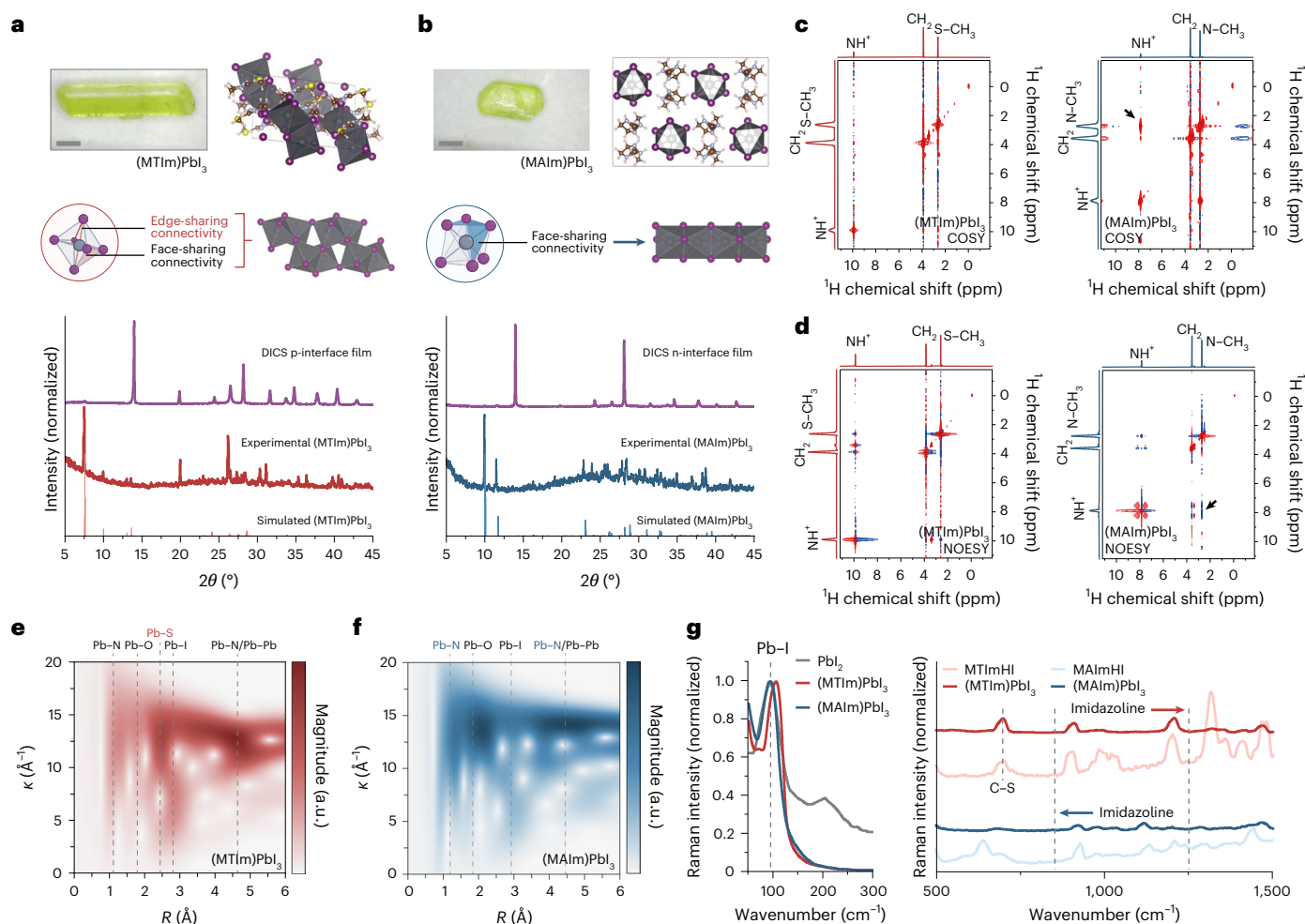


Fig. 1 | Crystal structures of 1D (MTIm)PbI₃ and (MAIm)PbI₃. **a, b**, Single-crystal photograph and structure of 1D (MTIm)PbI₃ (**a**) and (MAIm)PbI₃ (**b**) (top), and the corresponding experimental and simulated powder XRD patterns and the XRD pattern of the MAIm@n and MTIm@p perovskite films (bottom). Scale bars, 500 μ m. Atom colours: Pb, grey; I, purple; C, brown; S, yellow; N, blue-grey;

H, pink. **c, d**, Comparison of 2D NMR COSY (**c**) and NOESY (**d**) spectra of 1D (MTIm)PbI₃ and (MAIm)PbI₃ crystals. **e, f**, Wavelet transformations of the EXAFS images of 1D (MTIm)PbI₃ (**e**) and (MAIm)PbI₃ (**f**). The colour scale represents signal intensity, with white indicating low intensity and dark blue/red corresponding to high intensity. **g**, Raman spectra of PbI₂, (MTIm)PbI₃ and (MAIm)PbI₃ crystals.

However, achieving simultaneous and targeted energy-level regulation at both heterointerfaces remains highly challenging. The central difficulty lies in the limited understanding of how the interfacial behaviour of the products is governed by the intrinsic electronic structure of the modulator itself, given that the secondary perovskite phases typically adhere to similar electronic principles, where orbital contributions from metal–halide bonds dominate the band structure. This implies that molecular design guided by the bonding characteristics of heteroatoms and the resulting crystal frameworks could provide a rational route towards the coordinated control of interfacial charge selectivity and energy-level alignment.

Guided by this concept, we devised a dual-interface carrier-selective (DICS) strategy based on molecularly programmed octahedral connectivity. Two closely related imidazoline derivatives, 2-methylthio-2-imidazoline hydriodide (MTIm) and 2-methylamino-2-imidazoline hydriodide (MAIm), are designed by selectively incorporating sulfur and nitrogen atoms, respectively. Although they differ by only a single heteroatom (S– versus NH–), the secondary products exhibit contrasting work function (W_F) characteristics: (MTIm)PbI₃ shows a higher W_F , whereas (MAIm)PbI₃ shows a lower W_F . Mechanistic analysis reveals that the strong affinity of sulfur in MTIm to the perovskite lattice suppresses excessive reactivity of the imidazoline backbone, which expands the secondary phase through edge- and

face-sharing growth modes, shortens metal–halide bonds and ultimately favours hole-selective interfacial alignment. Consequently, benefiting from the precisely tailored charge-selective properties at both charge-collecting interfaces, PSCs based on the DICS architecture achieved a champion power conversion efficiency (PCE) of 27.61% (27.27% certified) and exhibited exceptional operational stability, retaining 97.8% and 87.4% of their initial efficiencies under ISOS L-1I and ISOS L-2I protocols—standardized ageing experiments established at the International Summit on Organic Photovoltaic Stability (ISOS)—after 2,000 h, respectively.

Crystal structures and interaction pathways of 1D (MTIm)PbI₃ and (MAIm)PbI₃

Single crystals were synthesized by reacting lead iodide (PbI₂) with the corresponding organic cations, MTIm and MAIm, to form the one-dimensional (1D) organic lead triiodides (MTIm)PbI₃ and (MAIm)PbI₃ (Fig. 1a, b), where their crystallographic parameters are summarized in Supplementary Tables 1 and 2, respectively. Single-crystal X-ray diffraction (XRD) analysis reveals that the two compounds crystallize in distinct space groups, *P21/c* for (MTIm)PbI₃ and *Pbca* for (MAIm)PbI₃, and adopt markedly different PbI₆ octahedral connectivity and framework geometry. Specifically, (MAIm)PbI₃ adopts exclusively face-sharing PbI₆ octahedra, forming a single-chain 1D

framework, whereas (MTIm)PbI₃ exhibits a denser configuration in which edge-sharing octahedra coexist with face-sharing connections, giving rise to two parallel edge- and face-sharing chains. The experimental diffraction patterns of the single crystals agree well with the simulated patterns. Distinct peaks corresponding to the (011) plane of (MTIm)PbI₃ at $2\theta = 7.52^\circ$ (d -spacing, 11.6 Å) and the (102) plane of (MAIm)PbI₃ at $2\theta = 9.94^\circ$ (d spacing, 8.8 Å) are clearly observed. It is noteworthy that the same diffraction signatures are also detected in the DICS-treated perovskite films in Fig. 1a,b, indicating faithful preservation of the corresponding 1D structural motifs in the film state.

Given this distinct structural divergence, we performed two-dimensional (2D) NMR spectroscopy, acquiring the COSY (correlation spectroscopy) and NOESY (nuclear Overhauser effect spectroscopy) spectra to probe the conformational responses and spatial correlations of the modulator molecules (MTIm and MAIm) on incorporation into the 1D lattices^{20,21}. In the COSY spectra (Fig. 1c), (MTIm)PbI₃ displays ¹H (proton) resonances at 9.90, 3.86 and 2.62 ppm, corresponding to the NH and CH₂ groups of the imidazoline backbone and the terminal methyl sulfur group, respectively. For (MAIm)PbI₃, the peaks at 7.86, 3.56 and 2.72 ppm are assigned to the NH and CH₂ groups of the backbone and the terminal methylamino group, respectively. A distinct difference is observed in the cross-signal patterns of the two samples. In (MAIm)PbI₃, pronounced cross-peaks connect the functional groups of the imidazoline ring with the methylamino substituent, whereas in (MTIm)PbI₃, no comparable additional cross-signals are observed. The same trend is observed when comparing the free modulators with their corresponding 1D products (Supplementary Fig. 1). These results indicate effective π -coupling within (MAIm)PbI₃, whereas (MTIm)PbI₃ exhibits minimal or negligible coupling. The implication is that the methyl sulfur terminal of MTIm hardly forms hydrogen bonds during 1D formation, whereas MAIm participates in directional hydrogen bonding, locking the imidazoline backbone into a more rigid configuration. The NOESY spectra provide complementary insights: whereas the proton chemical shifts remain nearly identical across all samples, the spatial correlations differ markedly (Fig. 1d). In (MTIm)PbI₃, strong cross-peaks span the three functional groups, indicating a more flexible arrangement²². However, these cross-signals are substantially attenuated in 1D (MAIm)PbI₃, indicating increased spatial separation and restricted conformational motion. The same distinction is observed when comparing the free modulators with their corresponding 1D materials (Supplementary Fig. 2). Solid-state ¹H-HMAS (magic-angle-spinning) NMR results also corroborate the differences in crystal structure (Supplementary Fig. 3 and Supplementary Note 1).

As such interaction pathways can ultimately dictate the framework geometry, the interactions extend beyond proton-mediated hydrogen bonding to include Pb-centred coordination, which plays a decisive role in the electronic states and energy band structure. To probe these interactions, we performed extended X-ray absorption fine-structure (EXAFS) measurements at the Pb L₃ edge on the two 1D perovskites. The Fourier-transformed EXAFS magnitude reveals that the first-shell Pb–I peak for (MAIm)PbI₃ appears at a higher apparent radial distance (R) value of 2.92 Å compared with 2.79 Å for (MTIm)PbI₃, indicating a longer average Pb–I bond distance in the MAIm-derived framework (Supplementary Fig. 4)^{23,24}. The wavelet transform EXAFS maps differentiate the local coordination environments: a distinct high- κ intensity feature that is attributable to Pb–S scattering is observed only in (MTIm)PbI₃, whereas Pb–N contributions are evident in both materials (Fig. 1e,f)^{25,26}. Notably, the Pb–N features in (MAIm)PbI₃ exhibit a stronger high- κ intensity than in (MTIm)PbI₃, implying a smaller Debye–Waller factor and a more ordered local environment around Pb for the MAIm-derived lattice²⁷, consistent with the NMR-inferred conformational locking.

Raman spectroscopy was used to corroborate these substituent-dependent lattice interactions (Fig. 1g). Compared with (MAIm)PbI₃, the Pb–I stretching mode in (MTIm)PbI₃ shows a

pronounced blueshift, reflecting increased bond covalency, reduced bond length and greater lattice rigidity²⁸. Concurrent blueshifts in the C–N–C ring-stretching and C–S modes further indicate strengthened π -conjugation and electronic coupling between the MTIm cation and the inorganic Pb–I framework. By contrast, (MAIm)PbI₃ shows only a slight redshift in the C–N–C stretching mode, indicating weaker participation of the heterocycle in the inorganic lattice, plausibly due to conformational locking induced by N–H $\cdots$ I hydrogen bonding.

X-ray photoelectron spectroscopy (XPS) analysis of pristine PbI₂ and the 1D crystalline products (MTIm)PbI₃ and (MAIm)PbI₃ further confirms the electronic anisotropy induced by substituent-dependent hybridization within the inorganic framework (Supplementary Fig. 5). The Pb 4f core levels in both compounds show similar negative shifts, indicating comparable local electronic perturbations. However, the I 3d core level in (MAIm)PbI₃ exhibits a larger negative shift, reflecting stronger modulation of electron density around the iodide due to N–H $\cdots$ I hydrogen bonding, consistent with the NMR and Raman results. By contrast, (MTIm)PbI₃ shows positive shifts in the N 1s and S 2p levels, providing evidence for coordination of the nitrogen and sulfur lone pairs with Pb²⁺, enhancing electronic coupling between the heterocycle and the Pb–I lattice. XPS of the perovskite films whose surfaces are modified with MTIm or MAIm further indicates stronger interactions between MAIm and the perovskite components (Supplementary Fig. 6). This indicates that substitution of the single heteroatom, S in (MTIm)PbI₃ versus N in (MAIm)PbI₃, is expected to redirect the interfacial interaction pathways when such 1D materials are positioned at the perovskite/charge-transport layer interface.

Crystal structure-dependent carrier-selective interfacial behaviour in 1D (MTIm)PbI₃ and (MAIm)PbI₃

To understand how substituent-dependent interaction motifs translate into modulation of the electronic structure, density functional theory (DFT) calculations were performed to analyse the band dispersion, projected density of states and Fermi-level positions of (MTIm)PbI₃ and (MAIm)PbI₃. The crystal structures used in the calculations were constructed directly from the experimentally resolved XRD structures of the 1D phases. To further probe the role of structural selectivity, we then hypothesized cross-assigning the lattice types to obtain face-sharing (MTIm)PbI₃ and face-/edge-sharing (MAIm)PbI₃ (Fig. 2a,b). Formation energy (ΔE_f) values were calculated for both the experimentally observed structures and these hypothesized swapped counterparts (Supplementary Table 3). Although all configurations are thermodynamically favourable, the values of their relative formation energy (ΔE_{rel}) in Fig. 2c indicate that the swapped structures are consistently less stable, with MTIm showing the more pronounced destabilization. This energetic hierarchy is consistent with experiments, where only the cation-preferred phases are obtained. The stability preference is further explained by differences in hydrogen bonding and coordination along the two linkage pathways (Supplementary Fig. 7, Supplementary Table 4 and Supplementary Note 2). The NH $\cdots$ I hydrogen bond in (MAIm)PbI₃ is stronger, enabling the cation to be more firmly locked in a rigid conformation, whereas the –S– in (MTIm)PbI₃ forms negligible hydrogen bonding, enabling greater flexibility. The larger internal empty volume in the face-/edge-sharing (MTIm)PbI₃ lattice also accommodates this increased conformational freedom (Supplementary Fig. 8 and Supplementary Table 5). These structural features agree well with the NMR results.

Notably, the double-chain geometry of (MTIm)PbI₃ also increases the iodide coordination number and strengthens Pb–I coupling. In the DFT models, (MTIm)PbI₃ exhibits a shorter average Pb–I bond length (3.22 Å) than (MAIm)PbI₃ (3.25 Å), indicating stronger orbital overlap and enhanced bonding interactions (Fig. 2d), in agreement with the EXAFS results. This is further supported by the integrated crystal orbital Hamilton population (ICOHP) analysis, where (MTIm)PbI₃

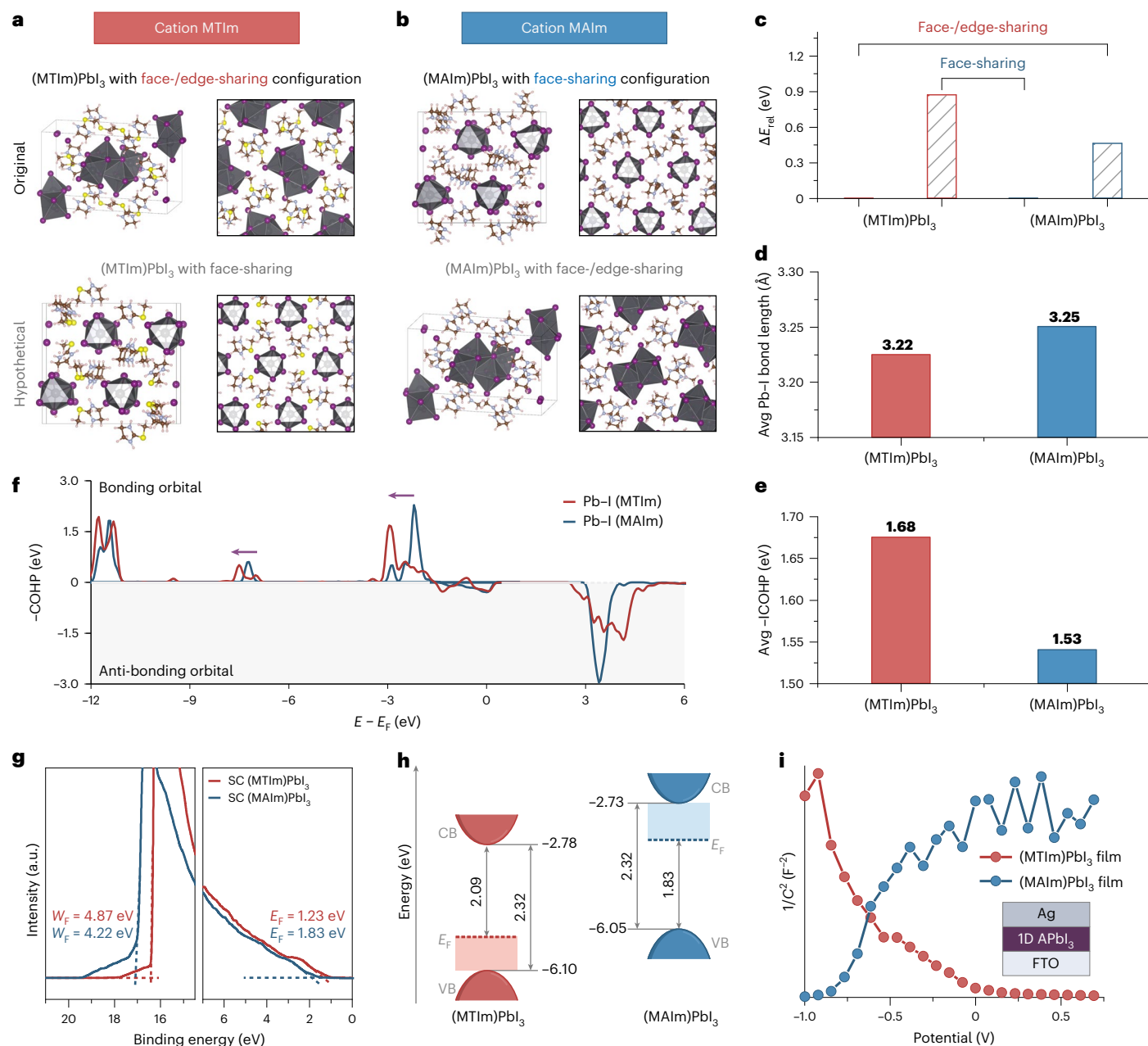


Fig. 2 | Crystal structure-dependent carrier-selective interfacial behaviour of 1D (MTIm)PbI₃ and (MAIm)PbI₃. **a, b**, Comparison of the 1D PbI₃ chain motifs for (MTIm)PbI₃ (**a**) and (MAIm)PbI₃ (**b**). The top row shows the experimentally observed structures: (MTIm)PbI₃ adopts a face-/edge-sharing double chain, whereas (MAIm)PbI₃ forms a face-sharing single chain. The bottom row displays the hypothetical 'cation exchange' models in which MTIm and MAIm are placed into each other's preferred structural frameworks. **c**, Relative formation energies

of (MTIm)PbI₃ and (MAIm)PbI₃ in the face-/edge-sharing versus face-sharing configurations. **d, e**, Comparison of the average (avg) Pb-I bond lengths (**d**) and ICOHP values (**e**) for all Pb-I bonds in (MTIm)PbI₃ and (MAIm)PbI₃. **f**, Electronic structure analysis of a Pb-I bond COHP. **g**, UPS spectra of single-crystal (SC) (MTIm)PbI₃ and (MAIm)PbI₃. **h**, Energy-level alignment of (MTIm)PbI₃ and (MAIm)PbI₃. CB, conduction band. **i**, Mott-Schottky plots of FTO/perovskite (PVSK)/Ag devices based on (MTIm)PbI₃ and (MAIm)PbI₃.

shows a more negative average ICOHP value (−1.68 eV) than (MAIm)PbI₃ (−1.53 eV) (Fig. 2e). These structural disparities directly reshape the electronic states through modified metal–halide orbital interactions. Energy-resolved crystal orbital Hamilton population (COHP) traces place the Pb–I bonding states of (MTIm)PbI₃ at deeper energy levels compared with those in (MAIm)PbI₃ (Fig. 2f), indicating more stabilized and delocalized orbital interactions. This enhanced coupling, reinforced by the denser Pb–I network, stabilizes the valence band (VB) and draws the Fermi level (E_F) downwards, favouring hole-selective interfacial alignment. By contrast, the comparatively weaker Pb–I bonding in (MAIm)PbI₃ leaves the bonding states shallower,

keeps E_F further from the VB edge and favours electron-selective alignment. DFT calculations performed on (MTIm)PbI₃ and (MAIm)PbI₃ further verify the energy levels (Supplementary Fig. 9). The DFT-derived work function of (MTIm)PbI₃ is 5.61 eV, which is higher than that of 5.40 eV for FAPbI₃ (where FA is formamidinium), whereas (MAIm)PbI₃ shows a lower W_F of 5.25 eV, consistent with the differentiated interfacial energy alignment. Strikingly, forcing the two cations into opposite structural motifs reverses their interfacial alignment tendencies. These results indicate that the Pb–I framework selectively adopts either a face-/edge-sharing or a purely face-sharing configuration depending on the cation. The Pb–I lattice governs the direction of the electronic

structure, whereas the cation mainly tunes the strength of this electronic response. Moreover, *ab initio* molecular dynamics simulations and phonon calculations indicate that the experimentally observed structures correspond to thermally robust representative local minima, rather than to a dynamically averaged ensemble of distinct configurations (Supplementary Figs. 10–12 and Supplementary Note 3).

The ultraviolet photoelectron spectroscopy (UPS) data in Fig. 2g, along with the extracted band energies summarized in Fig. 2h, show that (MTIm)PbI₃ exhibits a larger W_F of 4.87 eV and smaller energy offset between the Fermi level and the VB maximum ($E_F - E_{VBM}$) of 1.23 eV than (MAIm)PbI₃ ($W_F = 4.22$ eV; $E_F - E_{VBM} = 1.83$ eV), consistent with their respective interfacial band alignment characteristics (Fig. 2h, Supplementary Fig. 13 and Supplementary Table 6). Capacitance–voltage ($C-V$) measurements were carried out for Mott–Schottky analysis of the fluorine-doped tin oxide/1D perovskite/silver (FTO/(MTIm)PbI₃ or (MAIm)PbI₃/Ag) devices, and revealed a linear $1/C^2-V$ dependence in the depletion regime (Fig. 2i and Supplementary Note 4)^{29,30}. Notably, the (MTIm)PbI₃ film shows a negative slope that is characteristic of hole-selective band bending, whereas the (MAIm)PbI₃ film displays a positive slope that corresponds to electron-selective behaviour. Here the analysis is used primarily to probe the interfacial space-charge behaviour. The opposite responses reflect the shifts in the work function and band alignment variations that were observed via spectroscopy, arising from the distinct crystal symmetries of the two 1D phases that alter the Pb–I connectivity and electronic structure.

Dual-interface carrier-selective architecture design

Building on the distinct structural origins and carrier-selective character of (MTIm)PbI₃ and (MAIm)PbI₃, MTIm and MAIm were used as interfacial modulators at the charge-selective contacts of positive–intrinsic–negative (p–i–n) PSCs with a configuration of FTO/(4-(3,6-dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid/perovskite/fullerene/bathocuproine/Ag (FTO/Me-4PACz/perovskite/C₆₀/BCP/Ag) (Fig. 3a and Supplementary Fig. 14), where the treatments with MTIm or MAIm led to the *in situ* formation of 1D phases (Methods). Statistical analysis of the photovoltaic parameters reveals a pronounced dependence of device performance on the interfacial placement of the modulators relative to the charge-selective contacts (Fig. 3b, Supplementary Fig. 15 and Supplementary Table 7). The reference device (Ref.) delivers an average PCE of 24.08%. When MTIm is introduced at the n-type interface (MTIm@n) or MAIm at the p-type interface (MAIm@p), the performance improvements are marginal—the average PCE values increase slightly to 24.64 and 24.40%, respectively. By contrast, selectively matching the modulator with the corresponding contact significantly enhances the performance: MTIm at the p-type interface (MTIm@p) increases the average PCE to 26.58%, whereas MAIm at the n-type interface (MAIm@n) yields a PCE of 26.30%, accompanied by simultaneous improvements in the open-circuit voltage (V_{oc}) and fill factor (FF). Specifically, V_{oc} increases from 1.102 V in the reference device to 1.183 V for MTIm@p and 1.169 V for MAIm@n, whereas the FF improves from 83.04% in the reference device to 85.11 and 85.06% for MTIm@p and MAIm@n, respectively. These results establish MTIm and MAIm as effective hole- and electron-selective interfacial modulators, respectively.

Regarding the manipulation of both selective interfaces, when the two modulators are simultaneously applied but improperly positioned, that is MTIm@n + MAIm@p, the devices exhibit only minor improvements over that of the reference device, indicating that the beneficial effects are highly sensitive to the carrier-selective interfacial characteristics of the modulators and cannot be explained solely by defect passivation. By contrast, appropriate pairing of MTIm at the hole-transport layer (HTL) and MAIm at the electron-transport layer (ETL) (MTIm@p + MAIm@n) yields a further boost in device performance, delivering the highest average values ($V_{oc} = 1.212$ V, FF = 85.66% and PCE = 27.44%) as well as a champion PCE of 27.61% with

a short-circuit current density (J_{sc}) of 26.49 mA cm⁻², V_{oc} of 1.215 V and FF of 85.84% (Fig. 3c). The device also sustains a stable PCE of 27.16% (Supplementary Fig. 16). The J_{sc} value is well matched with the integrated external quantum efficiency (EQE) data (Supplementary Fig. 17). These results demonstrate that when properly aligned with the corresponding contacts, the MTIm- and MAIm-derived interfacial phases promote directional charge extraction, thereby suppressing interfacial recombination losses and contributing to the V_{oc} and FF. This strategy is further validated in wide-bandgap perovskite devices (Supplementary Figs. 18 and 19). Independent certification by the Chinese National Photovoltaic Industry Measurement and Testing Center verifies a reverse-scan PCE of 27.27% and a steady-state PCE of 27.19% after 300 s of maximum power point tracking for one of the DICS devices (Supplementary Figs. 20 and 21).

The interfacial structural evolution was examined using grazing-incidence wide-angle X-ray scattering (GIWAXS) at an incidence angle of 0.2°, which selectively probes the top ~10 nm of the perovskite films. A well-defined diffraction ring at $q = 1.00 \text{ \AA}^{-1}$ corresponds to the (100) plane of the three-dimensional (3D) perovskite phase (Fig. 3d and Supplementary Fig. 22). The treated films exhibit enhanced crystallinity and preferential out-of-plane orientation, as reflected by changes in the azimuthal intensity (Supplementary Fig. 23), consistent with the XRD results (Supplementary Fig. 24). GIXRD analysis also reveals a reduced lattice microstrain within the DICS film (Supplementary Fig. 25), indicating a more relaxed near-surface lattice environment. Together with the defect passivation and barrier effects of the 1D interfacial layer, this enhances thermal and moisture stability (Supplementary Fig. 26)^{31–34}. More importantly, the PbI₂ diffraction peak at $q = 0.90 \text{ \AA}^{-1}$ observed for the reference film is not detected from the modulator-treated samples, and instead, new features emerge. At the p-type interface treated with MTIm, an additional diffraction ring emerges at $q = 0.55 \text{ \AA}^{-1}$ which corresponds to a d -spacing of 11.6 Å, whereas an additional peak appeared at $q = 0.70 \text{ \AA}^{-1}$, corresponding to a d -spacing of 8.8 Å, is observed from the MAIm-treated samples at the n-type interface. These diffraction signatures are in good agreement with the single-crystal XRD patterns of (MTIm)PbI₃ and (MAIm)PbI₃, confirming the formation of 1D secondary phases at the intended interfaces.

Furthermore, cross-sectional high-resolution transmission electron microscopy (HRTEM) specimens prepared using focused ion beam milling provide direct visualization of these interlayers. In Fig. 3e distinct interlayers are seen compared with the surrounding 3D perovskite. At the p-type interface, an interlayer with a lattice d -spacing of 6.8 Å is identified, distinct from both the FTO substrate and the bulk perovskite. At the n-type interface, another interlayer exhibits a spacing of 9.4 Å, clearly differing from the adjacent amorphous C₆₀ and the 3D lattice (3.1 Å). Fast Fourier transforms validate the indexing, assigning them to the (120) plane of (MTIm)PbI₃ and the (200) plane of (MAIm)PbI₃, respectively (Supplementary Fig. 27). These findings confirm interface-localized formation of the 1D phases^{31–34}.

UPS measurements reveal W_F and band alignment modulation induced by the DICS strategy (Fig. 3f). At the p-type interface, W_F increases from 4.60 eV (reference) to 4.82 eV (DICS film), whereas the ionization potential (IP) decreases from 5.87 eV (reference) to 5.72 eV (DICS film). At the n-type interface, the W_F decreases from 4.89 to 4.85 eV along with a slight decrease in the IP from 5.92 to 5.90 eV for the reference and DICS film, respectively, consistent with the MAIm@n film (Supplementary Figs. 28 and 29). These shifts indicate that the DICS strategy effectively tailors interfacial energy-level alignment, promoting enhanced hole-selective extraction at the anode interface and electron-selective extraction at the cathode interface^{35–37}. With the perovskite absorber exhibiting a bandgap of 1.53 eV (Supplementary Fig. 30), further analysis of energy band alignment shows that the DICS strategy effectively compensates band-edge offsets at both interfaces (Fig. 3g). At the p-type interface, the offset

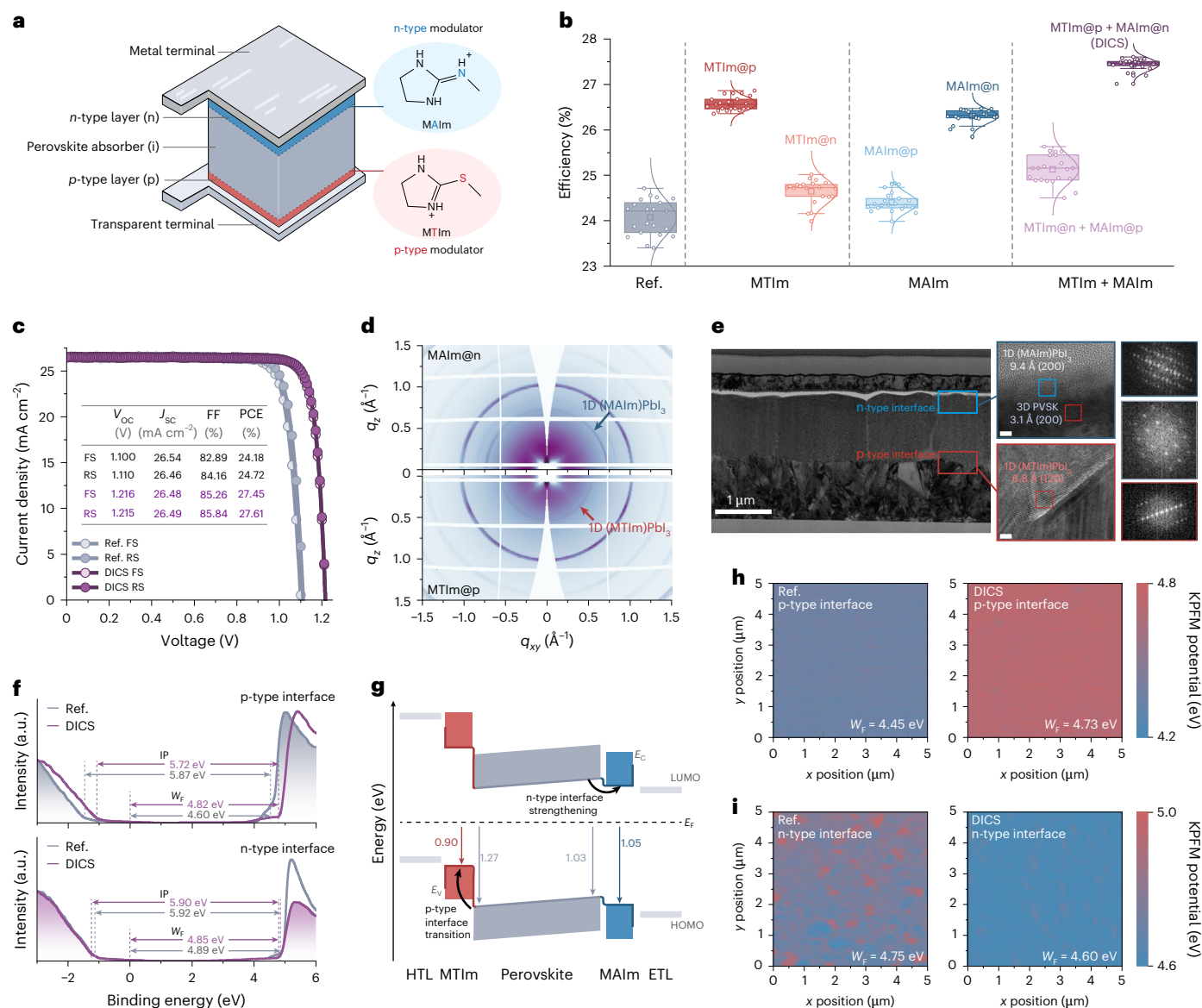


Fig. 3 | DICS architecture design and device performance. **a**, Schematic illustration of dual-interface modulation in the p-i-n architecture. **b**, Statistical distribution of PCEs for PSCs with different interfacial treatments, based on 21 individual devices. The box plot shows the minimum and maximum (whiskers), the mean (square symbols), the median (centre line) and the 25th–75th percentiles (box), with whiskers extending to 1.5 times the interquartile range. **c**, Current density–voltage (J – V) curves of the champion reference device and DICS p-i-n PSCs measured at laboratory scale under both reverse scan (RS) and forward scan (FS). **d**, GIWAXS snapshots of the n-type interface in the MAlm@n film and the p-type interface in the MTIm@p film. Dark purple and light colours correspond to high and low intensity signals, respectively. **e**, High-angle annular

dark-field TEM cross-sectional image (left), expansions (middle) and the corresponding fast Fourier transform patterns (right) of the DICS device. Scale bars (middle panels), 50 nm. **f**, UPS spectra of the p- and n-type interfaces in the reference and DICS perovskite films. E_F is defined as the zero of the x axis. The low-binding-energy region corresponds to the valence band onset relative to E_F , whereas the high-binding-energy region shows the secondary electron cut-off, shifted by the photon energy (21.22 eV) to determine the vacuum level. **g**, Energy band alignment diagrams of the reference and DICS PSCs. **h, i**, KPFM potential mappings of the p-type (h) and n-type (i) interfaces in the reference and DICS perovskite films.

between the perovskite VBM (E_v) and the HOMO (highest occupied molecular orbital) of the hole-selective layer decreases by 370 meV. At the n-type interface, the offset between the conduction band minimum (E_c) and the LUMO (lowest unoccupied molecular orbital) of the electron-selective layer is reduced by 20 meV. Notably, the pronounced reduction of the Schottky barrier at the p-type interface is consistent with the greater performance enhancement observed for MTIm@p in our device metrics. Kelvin probe force microscopy (KPFM) corroborates these findings. The work function at the p-type interface increases from 4.45 eV in the reference to 4.73 eV in the DICS film (Fig. 3h), whereas at the n-type interface it decreases from 4.75 to 4.60 eV,

respectively (Fig. 3i). The DICS films also show a more homogeneous distribution of the work function (Supplementary Figs. 31 and 32). These results establish the DICS strategy as a powerful approach for carrier-selective interfacial control, simultaneously compensating band-edge mismatches, minimizing Schottky barriers and reducing interfacial energy losses in PSCs^{38–40}.

Optoelectronic properties, upscaling and stability

Steady-state photoluminescence (SSPL) and time-resolved photoluminescence (TRPL) measurements of the perovskite/ETL and HTL/perovskite/ETL stacks show that the MTIm@p and DICS samples

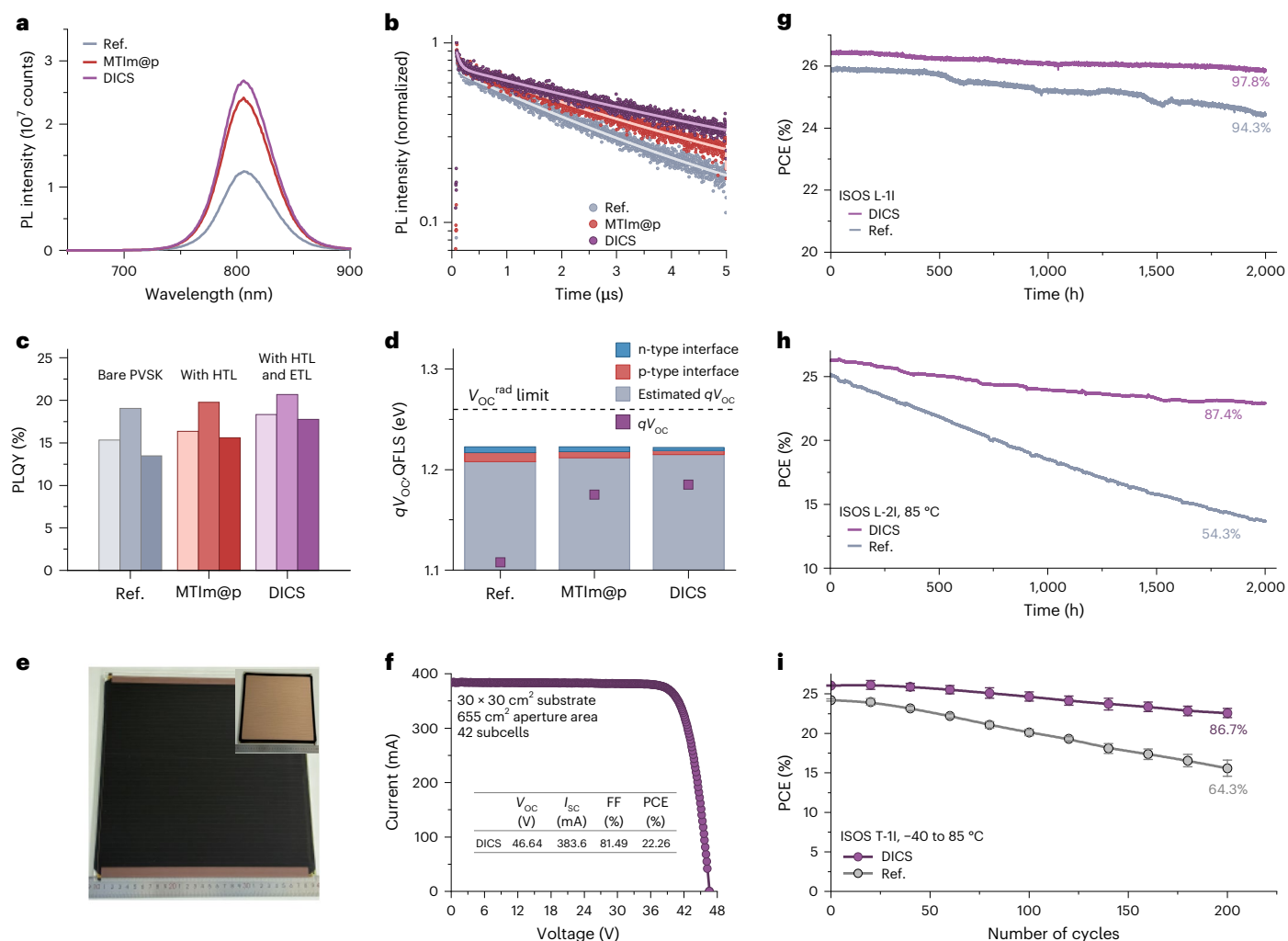


Fig. 4 | Photoelectronic performance and stability. **a, b**, SSPL (**a**) and TRPL (**b**) spectra of the reference, MTIm@p and DICS perovskite films deposited on the HTL. In **b**, the solid lines represent the fits to the PL decay curves. **c**, Comparison of the PLQY of the reference, MTIm@p and DICS perovskite samples. For each colour group, the three bars with increasing color intensity from light to dark represent the same perovskite film under three different conditions: bare perovskite, HTL/perovskite half-stack and HTL/perovskite/ C_{60} full-stack samples. **d**, Comparison of QFLS calculated from PLQY data. q , elementary charge; V_{OC}^{rad} , radiative V_{OC} . **e, f**, Photographs (**e**) and current-voltage (I - V) curves (**f**) of the DICS-modulated perovskite module with an aperture area of 655 cm^2 . I_{SC} , short-circuit current. In **e**, the main image and the inset show the front and back sides, respectively. **g**, PCE evolution of unencapsulated reference and DICS PSCs

under continuous tracking at the maximum power point following the ISOS L-1I protocol. The initial PCE values are 25.87% for the reference device and 26.43% for the DICS device. **h**, PCE evolution of the reference and DICS devices according to the ISOS L-2I protocol at maximum power point tracking under continuous 1 Sun illumination at 85 °C in an N_2 atmosphere. The initial PCE values are 25.15% for the reference device and 26.23% for the DICS device. **i**, Temperature cycling stability of the reference and DICS devices over 200 cycles (6 h per cycle) between -40 °C and 80 °C under the ISOS T-1I protocol. The initial PCE values are 24.23% for the reference device and 26.05% for the DICS device. Data are presented as the mean $\pm$ standard deviation. The sample size (n) denotes the number of devices tested, with $n = 3$ for all cases. Each curve represents the average values obtained from the statistical analysis of three independent samples for each parameter.

exhibit weaker PL emission and shorter lifetimes than the reference stack, indicative of more efficient carrier extraction across the interfaces (Supplementary Figs. 33 and Supplementary Tables 8 and 9)⁴¹. By contrast, in the HTL/perovskite half-stacks, both the PL intensity (Fig. 4a) and lifetime (Fig. 4b and Supplementary Table 10) increase relative to the reference, implying that MTIm suppresses interfacial recombination at the p-type contact. A similar enhancement is observed in neat films on a glass substrate, where the PL intensities for MTIm@p and DICS increase by nearly twofold and threefold, respectively, relative to the reference, consistent with the PL mapping results (Supplementary Fig. 34). The corresponding lifetimes extend from 2,746.6 ns for the reference to 3,294.1 and 3,548.6 ns for MTIm@p and DICS, respectively (Supplementary Fig. 34b and Supplementary Table 11), again indicating reduced non-radiative losses.

To quantify non-radiative recombination, the photoluminescence quantum yield (PLQY) was measured for bare perovskite, HTL/perovskite and HTL/perovskite/ETL stacks (Fig. 4c). The DICS sample yields a higher PLQY than the reference in both the bare perovskite and HTL/perovskite half-stack configurations, reflecting fewer interfacial non-radiative channels and more effective carrier separation. Even in the presence of an ETL, where rapid PL quenching typically reduces the PLQY, the DICS sample exhibits the smallest relative PLQY loss among the three samples, indicating suppressed defect-assisted recombination at the n-type contact. From the PLQY, we inferred the quasi-Fermi-level splitting (QFLS) to gauge the passivation effectiveness directly. The estimated interfacial voltage losses at the HTL/perovskite and perovskite/ETL stacks shrink to, respectively, 3.9 and 3.1 mV for DICS, and 9.0 and 5.7 mV for the reference (Fig. 4d), indicating deeper QFLS and a lower trap density at both interfaces^{42–45}.

Mott–Schottky analysis of full devices (Supplementary Fig. 35) further shows that the reference device exhibits pronounced curvature at low bias and a shallow high-bias slope, indicative of substantial interfacial defect density. These features are notably removed in the MTIm@p and DICS devices, which also show an increased built-in potential (V_{bi}), aligning with the enhanced QFLS and higher V_{oc} . Notably, the FF reaches 96.22% of the Shockley–Queisser limit (Supplementary Fig. 36), implying minimal transport losses across the interfaces. Space-charge-limited current measurements reveal substantial reductions in both electron and hole trap densities for the DICS device (Supplementary Fig. 37). Transient photovoltage and transient photocurrent experiments confirm slower recombination and faster carrier extraction (Supplementary Fig. 38). Light-intensity-dependent V_{oc} plots yield an ideality factor of 1.06 for DICS, lower than 1.22 for the reference, corroborating suppressed non-radiative recombination (Supplementary Fig. 39). Scalability is also assessed on 30×30 cm substrates. Forty-two-subcell modules with an aperture area of 655 cm^2 achieve a champion efficiency of 22.26%, with a V_{oc} of 46.64 V, a short-circuit current (I_{sc}) of 383.6 mA and an FF of 81.49% (Fig. 4e,f). These metrics position DICS as a manufacturing-relevant strategy rather than a laboratory-scale curiosity, highlighting its practical potential.

Device stability was evaluated under standard ISOS protocols. Under the ISOS L-II protocol, that is, unencapsulated devices tracked at maximum power under continuous illumination at room temperature in an inert atmosphere, the DICS device retains 97.8% of its initial PCE (26.43%) after 2,000 h, whereas the reference maintains 94.3% of its initial 25.87% (Fig. 4g). The improved retention is consistent with the reduced defect density and suppressed non-radiative recombination in the DICS devices. Thermal stability testing at 85°C based on the ISOS L-2I protocol shows that the DICS device maintains 87.4% of its initial PCE (26.23%) after 2,000 h, whereas the reference device declines to 54.3% of its initial 25.15% (Fig. 4h). Thermal cycling tests between -40 and 85°C based on the ISOS T-II protocol further demonstrated the robustness of the DICS architecture. The DICS device retains 86.7% of its initial PCE of 26.05% after 200 cycles (6 h per cycle), whereas the reference device is degraded to 64.3% of its initial 24.23% PCE (Fig. 4i). The improved durability is attributed to the DICS-induced interlayer phases that strengthen both interfaces and mitigate interfacial strain, consistent with the enhanced structural stability and lower tensile microstrain seen via XRD. It is probable that the enhanced performance and stability of the DICS-based devices arise from the combined effects of interfacial passivation, structural relaxation and improved band alignment. Although our results support these mechanisms, contributions from chemical passivation and other interfacial improvements cannot be ruled out.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-026-02722-3>.

References

- Kim, H.-S. et al. Lead iodide perovskite sensitized all-solid-state submicron thin film mesoscopic solar cell with efficiency exceeding 9%. *Sci. Rep.* **2**, 591 (2012).
- Xiong, Z. et al. Homogenized chlorine distribution for >27% power conversion efficiency in perovskite solar cells. *Science* **390**, 638–642 (2025).
- Zhang, Y. et al. Dynamical transformation of two-dimensional perovskites with alternating cations in the interlayer space for high-performance photovoltaics. *J. Am. Chem. Soc.* **141**, 2684–2694 (2019).
- Liang, Z. et al. Homogenizing out-of-plane cation composition in perovskite solar cells. *Nature* **624**, 557–563 (2023).
- Cho, S.-H. et al. Solution-phase PbI_2 coordination controls perovskite film formation and photovoltaic performance. *ACS Energy Lett.* **10**, 3931–3940 (2025).
- Canil, L. et al. Tuning halide perovskite energy levels. *Energy Environ. Sci.* **14**, 1429–1438 (2021).
- Peng, Y. et al. Enlarging moment and regulating orientation of buried interfacial dipole for efficient inverted perovskite solar cells. *Nat. Commun.* **16**, 1252 (2025).
- Zhang, C. et al. Work function tuning of a weak adhesion homojunction for stable perovskite solar cells. *Joule* **8**, 1394–1411 (2024).
- Zhang, J., Sun, Y. & Yu, H. Reducing energy loss via adjusting the anode work function and perovskite layer morphology for the efficient and stable hole transporting layer-free perovskite solar cells. *Chem. Eng. J.* **431**, 133948 (2022).
- Ren, B. et al. A universal p-type heterointerface for inverted perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202507950 (2025).
- Liu, S. et al. Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536–542 (2024).
- Zhou, C. et al. Dual interfacial modification with 1D perovskite for self-assembled monolayer based inverted perovskite solar cells. *Nano Energy* **128**, 109811 (2024).
- Jeon, Y.-S. & Park, N.-G. Nickel oxide thin films formed from the pH-adjusted solutions for inverted perovskite solar cells. *ACS Energy Lett.* **10**, 1266–1274 (2025).
- Liu, G. et al. Mixed-phase low-dimensional perovskite-assisted interfacial lead directional management for stable perovskite solar cells with efficiency over 24%. *ACS Energy Lett.* **6**, 4395–4404 (2021).
- Xue, J. et al. Reconfiguring the band-edge states of photovoltaic perovskites by conjugated organic cations. *Science* **371**, 636–640 (2021).
- Li, M., Huang, P. & Zhong, H. Current understanding of band-edge properties of halide perovskites: Urbach tail, Rashba splitting, and exciton binding energy. *J. Phys. Chem. Lett.* **14**, 1592–1603 (2023).
- Daboczi, M. et al. Optimal interfacial band bending achieved by fine energy level tuning in mixed-halide perovskite solar cells. *ACS Energy Lett.* **6**, 3970–3981 (2021).
- Kutkan, S. et al. Impact of the organic cation on the band-edge emission of two-dimensional lead–bromide perovskites. *Nanoscale* **15**, 12880–12888 (2023).
- Li, X. et al. Constructing heterojunctions by surface sulfidation for efficient inverted perovskite solar cells. *Science* **375**, 434–437 (2022).
- Monika, Verma, A., Tiwari, M. K., Show, B. & Saha, S. Modulation of weak interactions in structural isomers: positional isomeric effects on crystal packing and physical properties and solid-state thin-film fabrication. *ACS Omega* **5**, 448–459 (2019).
- Zeng, M. et al. Tailored supramolecular interface enables efficient and stable tin halide perovskite photovoltaics. *ACS Energy Lett.* **10**, 1357–1365 (2025).
- Dahlman, C. J., Kubicki, D. J. & Reddy, G. N. M. Interfaces in metal halide perovskites probed by solid-state NMR spectroscopy. *J. Mater. Chem. A* **9**, 19206–19244 (2021).
- Hui, W. et al. Stabilizing black-phase formamidinium perovskite formation at room temperature and high humidity. *Science* **371**, 1359–1364 (2021).
- Kang, D.-H., Park, Y.-J., Jeon, Y.-S. & Park, N.-G. Extended X-ray absorption fine structure (EXAFS) of FAPbI_3 for understanding local structure–stability relation in perovskite solar cells. *J. Energy Chem.* **67**, 549–554 (2022).
- Zhang, B. et al. Inhibiting ion migration and stabilizing crystal-phase in halide perovskite via directly incorporated fluoride anion. *Angew. Chem. Int. Ed.* **137**, e202413550 (2025).

26. Wen, Y. et al. Amorphous (lysine)₂PbI₂ layer enhanced perovskite photovoltaics. *Nat. Commun.* **15**, 7085 (2024).
27. Sanson, A. EXAFS spectroscopy: a powerful tool for the study of local vibrational dynamics. *Microstructures* **1**, 2021004 (2021).
28. Ibaceta-Jaña, J. et al. Vibrational dynamics in lead halide hybrid perovskites investigated by Raman spectroscopy. *Phys. Chem. Chem. Phys.* **22**, 5604–5614 (2020).
29. Zheng, J. et al. Incorporation of CoO nanoparticles in 3D marigold flower-like hierarchical architecture MnCo₂O₄ for highly boosting solar light photo-oxidation and reduction ability. *App. Catal. B* **237**, 1–8 (2018).
30. Yang, Y. et al. Inverted perovskite solar cells with over 2,000 h operational stability at 85 °C using fixed charge passivation. *Nat. Energy* **9**, 37–46 (2024).
31. Ji, R. et al. Spontaneous formation of 1D/3D perovskite heterojunctions for efficient inverted perovskite solar cells. *Adv. Energy Mater.* **14**, 2304126 (2024).
32. Wei, N. et al. Post-treatment-free dual-interface passivation via facile 1D/3D perovskite heterojunction construction. *JACS Au* **3**, 3324–3332 (2023).
33. Chen, R. et al. Conformal imidazolium 1D perovskite capping layer stabilized 3D perovskite films for efficient solar modules. *Adv. Sci.* **9**, 2204017 (2022).
34. Jiao, H. et al. Perovskite grain wrapping by converting interfaces and grain boundaries into robust and water-insoluble low-dimensional perovskites. *Sci. Adv.* **8**, eabq4524 (2022).
35. Xiong, W. et al. Controllable p- and n-type behaviours in emissive perovskite semiconductors. *Nature* **633**, 344–350 (2024).
36. Xiong, S. et al. Direct observation on p- to n-type transformation of perovskite surface region during defect passivation driving high photovoltaic efficiency. *Joule* **5**, 467–480 (2021).
37. Zheng, Y. et al. Towards 26% efficiency in inverted perovskite solar cells via interfacial flipped band bending and suppressed deep-level traps. *Energy Environ. Sci.* **17**, 1153–1162 (2024).
38. Al-Ashouri, A. et al. Monolithic perovskite/silicon tandem solar cell with >29% efficiency by enhanced hole extraction. *Science* **370**, 1300–1309 (2020).
39. Cacovich, S. et al. Imaging and quantifying non-radiative losses at 23% efficient inverted perovskite solar cells interfaces. *Nat. Commun.* **13**, 2868 (2022).
40. Xu, H. et al. Constructing robust heterointerfaces for carrier viaduct via interfacial molecular bridges enables efficient and stable inverted perovskite solar cells. *Energy Environ. Sci.* **16**, 5792–5804 (2023).
41. Ye, F. et al. Overcoming C₆₀-induced interfacial recombination in inverted perovskite solar cells by electron-transporting carborane. *Nat. Commun.* **13**, 7454 (2022).
42. Warby, J. et al. Mismatch of quasi-Fermi level splitting and V_{oc} in perovskite solar cells. *Adv. Energy Mater.* **13**, 2303135 (2023).
43. Caprioglio, P. et al. Open-circuit and short-circuit loss management in wide-gap perovskite p-i-n solar cells. *Nat. Commun.* **14**, 932 (2023).
44. Duan, C. et al. Scalable fabrication of wide-bandgap perovskites using green solvents for tandem solar cells. *Nat. Energy* **10**, 318–328 (2025).
45. Sarritzu, V. et al. Optical determination of Shockley–Read–Hall and interface recombination currents in hybrid perovskites. *Sci. Rep.* **7**, 44629 (2017).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

Methods

Materials

MAIm (99%), MTIm (99%), caesium iodide (CsI, 99.999%), dimethylformamide (DMF, 99.8%), chlorobenzene (99.8%), isopropanol (IPA, 99.9%), dimethyl sulfoxide (DMSO, 99.9%) and *N*-methylpyrrolidine (NMP, 98%) were purchased from Sigma-Aldrich. PbI_2 (99.99%), Me-4PACz (>99.0%) and MeO-2PACz (>98.0%) were purchased from TCI Chemicals. Methylammonium chloride (MAcI, 99.99%) was purchased from Greatcell Solar Materials. Formamidinium iodide (FAI) was synthesized by reacting formamidinium acetate (99%, Sigma-Aldrich) with hydriodic acid (HI; 57 wt% diluted in H_2O , Sigma-Aldrich) in an ice bath. C_{60} was purchased from Nano-C. All chemicals were used as received without further purification.

Perovskite precursor solution

A 1.6 M $\text{FA}_{0.95}\text{Cs}_{0.05}\text{PbI}_3$ precursor solution was prepared by mixing stoichiometric amounts of FAI, CsI and PbI_2 in a DMF/DMSO mixed solvent (DMF:DMSO = 4:1, v/v), in which an additional 8 mol% PbI_2 and 10 mol% MAcI were dissolved. Subsequently, 0.3 mg ml^{-1} (phenylsulfonyl)pyrrole was added⁴ and the solution was stirred for another 2 h, followed by filtration before use.

Device fabrication

DICS film preparation. For the p-type interlayer treatment, an MTIm/IPA solution (0.8 mg ml^{-1}) was dynamically spin-coated onto the substrate at 4,000 rpm for 30 s and subsequently annealed at 100 °C for 5 min. The filtered perovskite precursor was then deposited via spin-coating at 1,000 rpm for 10 s and 5,000 rpm for 30 s, with chlorobenzene (300 μl) added dropwise as an anti-solvent at 13 s during the spin process, followed by annealing at 100 °C for 45 min. For the n-type interlayer treatment, an MAIm/IPA solution (1 mg ml^{-1}) was dynamically spin-coated onto the perovskite layer at 3,000 rpm for 30 s and then annealed at 100 °C for 5 min.

Fabrication of small-area narrow-bandgap devices. Glass/FTO substrates were sequentially cleaned via ultrasonication in detergent, deionized water, acetone and IPA for 20 min each. Before deposition of the hole-transport layer, the substrates were treated with oxygen plasma for 5 min, and immediately transferred into a glovebox with an N_2 atmosphere. A solution of Me-4PACz (0.6 mg ml^{-1} in ethanol) was spin-coated at 5,000 rpm for 30 s and annealed at 100 °C for 10 min. The perovskite films were subsequently deposited by spin coating using the aforementioned procedure. Finally, C_{60} (25 nm, 0.1 $\text{\AA} \text{ s}^{-1}$), BCP (7 nm, 0.1 $\text{\AA} \text{ s}^{-1}$) and Ag (100 nm, 0.2 $\text{\AA} \text{ s}^{-1}$) were successively thermally evaporated in a vacuum chamber (7×10^{-5} Pa) to complete fabrication of the device.

Fabrication of wide-bandgap devices. The clean FTO substrates were plasma-treated for 15 min. A solution of nickel oxides (NiO_x) nanoparticles (10 mg ml^{-1} in deionized water) was spin-coated at 4,000 rpm for 30 s and annealed at 150 °C for 20 min before being transferred to a glovebox. A layer of MeO-2PACz (10 mg ml^{-1} in IPA) was deposited onto the NiO_x surface at 3,000 rpm for 30 s and annealed at 120 °C for 10 min. The perovskite layer, $\text{Cs}_{0.05}(\text{FA}_{0.77}\text{MA}_{0.23})_{0.95}\text{Pb}(\text{I}_{0.77}\text{Br}_{0.23})_3$, was deposited via spin-coating the precursor solution (DMF:DMSO = 4:1, v/v) at 1,000 rpm for 5 s and 4,000 rpm for 30 s. During the final 10 s, ethyl acetate (500 μl) was added dropwise as an anti-solvent, followed by annealing at 120 °C for 20 min. The interfacial treatment was the same as that used for the narrow-bandgap devices. Layers of PC_{61}BM (a soluble derivative of C_{60} ; 20 mg ml^{-1} in chlorobenzene) and BCP (1 mg ml^{-1} in IPA) were sequentially spin-coated at 2,000 and 5,000 rpm for 30 s, respectively. Finally, a 90-nm-thick Ag electrode was added via thermal evaporation.

Fabrication of slot-die-printed perovskite submodules. For the 30×30 cm perovskite solar module, the conductive glass substrates were ultrasonically cleaned as described above. P1 scribe lines (20 μm

wide) were ablated on the substrates using a 1,064-nm nanosecond laser. A 20-nm-thick NiO_x layer was deposited via direct-current sputtering. A Me-4PACz solution (0.3 mg ml^{-1}) was slot-die printed at a speed of 15 mm s^{-1} and annealed for 10 min. The MTIm layer (0.3 mg ml^{-1}) was blade-coated with a 120- μm gap at 8 mm s^{-1} and annealed for 5 min. A 1.1 M perovskite precursor ink (DMF:NMP = 7:1 v/v) was slot-die coated at 15 mm s^{-1} with a pump rate of 20 $\mu\text{l} \text{ s}^{-1}$ and a head gap of 60 μm . The wet film was immediately transferred into a chamber evacuated to 10 Pa for 60 s, then annealed at 100 °C for 30 min in air. Subsequently, an MAIm layer (0.5 mg ml^{-1}) was blade-coated under the same conditions and annealed for 5 min. Electron-transport layers consisting of 20-nm-thick C_{60} (via thermal evaporation) and 17-nm-thick SnO_x (via atomic layer deposition) were sequentially deposited. P2 lines (40 μm wide) were scribed near the P1 lines using a laser. Finally, 20-nm-thick indium tin oxide and 80-nm-thick Cu electrodes were deposited, followed by P3 (30 μm wide) and P4 scribe lines using a 532-nm picosecond laser. Edge isolation was achieved via P4 patterning using a 1,064-nm nanosecond laser.

Device characterization

Photovoltaic performance. The J – V characteristics were measured using a Keithley 2400 source meter under simulated AM 1.5G solar illumination (100 mW cm^{-2}) provided by a xenon lamp-based solar simulator (Enlitech SS-F5-3A, Class AAA). The light intensity was calibrated using a certified silicon reference cell (NREL-traceable, SRC-2020). A metal aperture mask defining an active area of 0.06 cm^2 was used during the measurements. The voltage was swept from –0.2 to 1.3 V at a rate of 0.2 V s^{-1} with a voltage step of 20 mV. EQE spectra were recorded in alternating-current mode using a xenon lamp source (Newport TLS260-300X) over a wavelength range of 300–1,000 nm.

Stability. A calibrated light-emitting diode solar simulator (100 mW cm^{-2}) was used for stability testing with maximum power point tracking under N_2 . For the thermal stress and cycling tests, BCP/Ag was replaced with SnO_2/Au to improve the thermal stability. Under the ISOS-L21 protocol, devices were aged at 85 °C and periodically characterized via J – V scans after cooling. Thermal cycling was performed in a custom chamber with J – V measurements after each temperature stabilization cycle.

Structural characterization

XRD and GIXRD data of the perovskite films were obtained using a Rigaku SmartLab diffractometer, with Cu $\text{K}\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$). GIWAXS ($\lambda = 1.24 \text{ \AA}$; incidence angle, 0.10°) and Pb L_3 -edge EXAFS (200 mA at 3.5 GeV; incidence angle, 0.20°) were performed at beamlines BL17B1 and BL13SSW of the Shanghai Synchrotron Radiation Facility. Cross-sectional HRTEM adopts FTO/Me-4PACz/perovskite/ $\text{PC}_{61}\text{BM}/\text{Cu}$ architecture. The Cu thickness here was 200 nm, to protect the perovskite film from milling damage. Pt and C layers were deposited before thinning using a focused ion beam device (ThermoFisher, Helios 5 CX).

Single crystals of 1D (MTIm) PbI_3 and (MAIm) PbI_3 were obtained by dissolving equimolar amounts of PbI_2 and MTIm and MAIm in a mixed solvent of HI and H_3PO_2 (at a volume ratio of 1:0.2), with stirring at 100 °C for 1 h, before allowing the mixture to cool slowly to room temperature. A suitable crystal was selected and mounted on a Bruker APEX-II CCD diffractometer. Data were obtained using monochromatic Ga $\text{K}\alpha$ radiation at $T = 100$ or 200 K. Data reduction, scaling and absorption corrections were performed using the SAINT v8.38A software package (Bruker, 2013). The structure was solved with the ShelXT structure solution program using the intrinsic phasing solution method; Olex2 was used as the graphical interface. The model was refined with version 2016/6 of ShelXL using least squares minimization. The resulting CIF (crystallographic information file) and structure factors are validated using the checkcif test, and no alert A or B is reported.

The 2D NMR COSY and NOESY spectra were obtained using a Quantum-Plus AS600 instrument (Zhongke Oxford). 1D (MTIm)PbI₃ and (MAIm)PbI₃ were respectively dissolved in DMSO. 2D MAS NMR measurements were conducted on solid samples using an Ascend 800 AVANCE NEO instrument (Bruker).

Characterization of film photoelectric and charge-carrier properties

SSPL and TRPL spectra were measured using an FLS1000 spectrometer (Edinburgh Instruments) with 405-nm beam source. The PL decay curves were fitted with biexponential components. The optical absorbance was measured using an ultraviolet-visible spectrometer (Lambda 45, PerkinElmer). XPS was carried out using an ESCALAB250 instrument with a monochromatized Al source ($h\nu = 1,486.6$ eV). The samples were prepared by spin-coating the perovskite layer on 0.5×0.5 cm Si substrates. KPFM was carried out using an atomic force microscope (NX10, Park Systems) with a Pt-coated conductive tip under an N₂ atmosphere. The Mott–Schottky plots of 1D (MTIm)PbI₃ and (MAIm)PbI₃ were measured at a frequency of 2,000 Hz and within the applied bias range of -1.0 to 1.0 V. Raman spectra were recorded using an XperRAM-S567 system with a 532-nm excitation laser and an exposure time of 3 s.

Computational details

All DFT calculations were performed using the Vienna ab initio simulation package (VASP v5.4.4) for periodic systems⁴⁶. For the VASP calculations, the projector augmented-wave method was used, and exchange–correlation interactions were treated using the Perdew–Burke–Ernzerhof functional under the generalized gradient approximation^{47,48}. Integration in the Brillouin zone was performed on the basis of the Monkhorst–Pack scheme using $2 \times 2 \times 2$ k-point meshes in each primitive lattice vector of the reciprocal space for geometry optimization of the bulk structures of APbI₃ (where A is MTIm or MAIm), including the corresponding reversed phases⁴⁹. We used the DFT-D3 dispersion correction method to reflect the correlation of the non-bonding interactions in the 1D perovskite structure⁵⁰. Lattice constants and internal atomic positions were fully optimized using a plane-wave cut-off energy of 500 eV until the residual forces were less than 0.04 eV Å⁻¹. In addition, the COHP analysis was used to understand chemical bonds by dividing the electronic structure into bonding and anti-bonding contributions using the LOBSTER (local orbital basis suite towards electronic-structure reconstruction) code based on the pbeVasFit2015 basis set^{51–53}. Moreover, as a descriptor for numerically evaluating the chemical bond strength, we considered the ICOHP value up to the Fermi level. To examine structural variations under thermal conditions, ab initio molecular dynamics simulations were performed in the NVT ensemble (constant number of particles, volume and temperature) using a Nosé–Hoover thermostat at 300 K for 6 ps with a time step of 1 fs (refs. 47,54).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

References

- Kresse, G. & Hafner, J. Ab initio molecular dynamics for open-shell transition metals. *Phys. Rev. B* **48**, 13115–13118 (1993).
- Kresse, G. & Hafner, J. Ab initio molecular-dynamics simulation of the liquid-metal–amorphous-semiconductor transition in germanium. *Phys. Rev. B* **49**, 14251–14269 (1994).
- Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
- Monkhorst, H. J. & Pack, J. D. Special points for Brillouin-zone integrations. *Phys. Rev. B* **13**, 5188–5192 (1976).
- Grimme, S., Antony, J., Ehrlich, S. & Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *J. Chem. Phys.* **132**, 154104 (2010).
- Dronskowski, R. & Blöchl, P. E. Crystal orbital Hamilton populations (COHP): energy-resolved visualization of chemical bonding in solids based on density-functional calculations. *J. Phys. Chem.* **97**, 8617–8624 (1993).
- Deringer, V. L., Tchougréeff, A. L. & Dronskowski, R. Crystal orbital Hamilton population (COHP) analysis as projected from plane-wave basis sets. *J. Phys. Chem.* **115**, 5461–5466 (2011).
- Maintz, S., Deringer, V. L., Tchougréeff, A. L. & Dronskowski, R. LOBSTER: a tool to extract chemical bonding from plane-wave based DFT. *J. Comput. Chem.* **37**, 1030–1035 (2016).
- Hoover, W. G. Canonical dynamics: equilibrium phase-space distributions. *Phys. Rev. A* **31**, 1695–1697 (1985).

Acknowledgements

This work was supported by National Research Foundation of Korea (NRF) grants funded by the Korean government (MSIT and MOE) under contracts RS-2026-25501632 (NRL 2.0), NRF-2021R1A3B1076723 (Research Leader Program), RS-2025-02316700 (Carbon-free Energy Core Technology Program), RS-2023-00259096 (GRDC Cooperative Hub) and the Ministry of Trade, Industry and Energy (MOTIE) of Korea (P0022336). We thank the Shanghai Synchrotron Radiation Facility for use of the BL17B1 and BL13SSW beamlines (cstr.cn/31124.02.SSRF, BL13SSW) and for support with the XAFS experiments.

Author contributions

Y.Z., Z.L. and N.-G.P. conceived the study. N.-G.P. supervised the project. Y.Z. and Z.L. fabricated the perovskite devices, performed the photovoltaic characterizations and analysed the experimental data. Y.Z. and Z.L. wrote the paper. N.-G.P. revised and edited the paper. G.F. performed the KPFM measurements. B.L. and H.Z. assisted with the EXAFS analysis. X.C. conducted and analysed the GIWAXS measurements under the supervision of K.Z. The XRD measurements were performed by S.-U.L. The DFT calculations were carried out by S.C.C. and J.H.S. under the supervision of S.U.L. Analysis of the NMR data was assisted by S.-H.C. and S.L. All authors discussed the results and contributed to the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41563-026-02722-3>.

Correspondence and requests for materials should be addressed to Sang Uck Lee, Xu Pan or Nam-Gyu Park.

Peer review information *Nature Materials* thanks Milos Dubajic and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on Nature Research policies, including our [data availability policy](#), see [Authors & Referees](#).

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

Area of the tested solar cells	☒ Yes ☐ No	Report the area of the tested solar cells. 0.06 cm ²
Method used to determine the device area	☒ Yes ☐ No	Provide a description of the method and state where this information can be found in the text. Metal aperture mask
2. Current-voltage characterization		
Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	The certified J-V in both forward and backward direction in Supplementary Fig.20
Voltage scan conditions	☒ Yes ☐ No	Provide a description of the measurement conditions (e.g. scan direction, speed, dwell times).
Test environment	☒ Yes ☐ No	Provide a description of the test conditions (e.g. characterization temperature, atmosphere, humidity). Room temperature and nitrogen gloved box
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	Provide a description of the protocol.
Stability of the J-V characteristic	☒ Yes ☐ No	Provide a description of the method used. The stability of the J-V characteristic can be verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 5 for details. The stable power output PCE at MPP condition in Supplementary Fig.16 and Supplementary Fig.21
Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	Provide a description of hysteresis or any other unusual behaviour observed during the characterization. Slight hysteresis
Related experimental data	☒ Yes ☐ No	Provide a description of the related experimental data. Supplementary Fig.20
External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	Provide a description of the technique used. Supplementary Fig.17
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	Fig.3c

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. See details in Method.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Provide a description of the light source and reference cell or sensor.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Identify the independent certification laboratory. Our solar simulator was calibrated by Si reference cell. Certificated by Newport.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements. Supplementary Fig.21
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Report the size of the mask/aperture. 0.06 cm2
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Identify the independent certification laboratory. Supplementary Fig.20,21
A copy of any certificate(s)	☒ Yes ☐ No	Certificate copies should be provided in the Supplementary information. Please state the supplementary item number. Supplementary Fig.20,21
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Report how many solar cells have been tested, specifying the number of individual substrates. 21
Statistical analysis of the device performance	☒ Yes ☐ No	Fig.3b, Supplementary Fig.15
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Provide a description of the type of analysis, bias conditions and environmental conditions (e.g. illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature, bias) for each long-term stability analysis carried out; see ref. 7 and 8 for details. Continous MPP tracking under continous 1 sun LED illumination (Fig.4g); Thermal stability at MPP tracking under continuous 1 sun illumination at 85 oC in N2 glovebox (Fig.4g); Temperature cycling stability for 1200 h between -40 to 80 oC (Fig.4i)